THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 4: **Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which one of these pairs contains only chemical changes? 1

(a) burning of coal, ripening of fruit **(b)** cutting paper, tearing cloth
(c) dissolving sugar, freezing water **(d)** melting wax, boiling water

2. In the experiment with three nails, the nail in dry air does not rust because: 1

(a) no moisture **(b)** no iron **(c)** no oxygen **(d)** no sunlight

3. Which statement about crystallisation is true? 1

(a) crystals form by burning
(b) crystals form from a gas always
(c) crystals form by sieving
(d) crystals form when a hot saturated solution cools

4. Which one of these pairs contains only physical changes? 1

(a) digestion, photosynthesis **(b)** cooking, curdling
(c) burning wood, rusting **(d)** freezing of water, evaporation of water

5. In the experiment, the nail in boiled water with oil does not rust because: 1

(a) too much oil **(b)** no iron
(c) no water **(d)** no air reaches it

6. In making crystals, why is the solution filtered while hot? 1

(a) to remove all water **(b)** to make it blue
(c) to make it solid **(d)** to remove insoluble impurities

7. Which of the following changes cannot be reversed? 1

(a) folding of paper **(b)** freezing of water
(c) frying of an egg **(d)** melting of butter

8. In the nail experiment, the nail in tap water exposed to air rusts because: 1

(a) both air and water are present **(b)** sunlight is present
(c) only air is present **(d)** only water is present

9. Why do we leave the solution undisturbed to get large crystals? 1

(a) shaking gives bigger crystals **(b)** light gives crystals
(c) heat gives crystals **(d)** slow cooling gives bigger crystals

10. A student dissolves salt in water and then evaporates the water. The salt is recovered. This shows: 1

(a) salt disappears **(b)** dissolving is a physical change

(c) salt is a gas **(d)** dissolving is chemical

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the burning of a candle called both a physical and a chemical change? 2

12. Why are crystals formed when a hot saturated solution is cooled? 2

13. Explain how a galvanised bucket resists rusting. 2

14. Give two reasons why rusting is a chemical change. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. You are given a piece of iron nail and a copper sulphate solution. Predict the changes and say whether the change is physical or chemical. 3

16. Why is photosynthesis a chemical change? 3

17. Compare cooking of rice with melting of butter. 3

OR

Describe the observations and conclusions when a few drops of lemon juice are added to a baking soda solution. 3

18. Why are sweets packed in airtight containers to prevent chemical changes? 3

OR

Explain two ways of preventing rust in a farmer's iron plough. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Plan an investigation to find the effect of salt water and plain water on rusting of iron nails, with the set-up, observations and conclusion. 5

OR

Discuss the harmful and useful chemical changes with three examples each. 5

20. Explain the changes that occur in a car's exhaust, the engine, the paint and the iron body, and classify them. 5

OR

Explain how the change of colour of a cut apple can be slowed down. 5

END OF QUESTION PAPER — Total: 40 Marks

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